EULOGIO "AMANG" RODRIGUEZ INSTITUTE OF SCIENCE & TECHNOLOGY

Campus Health Services • Clinic Appointment & Triage System

DATABASE DATA DICTIONARY & COMPLETE SCHEMA SPECIFICATION

Database Engine: SQLite (node:sqlite)
Database File: earist_clinic.db
Schema Version: 1.2 (With Data Type Lengths)
Generated: October 22, 2026

TOTAL TABLES: 6 Relational Tables	**DATA TYPE LENGTHS:** Explicit Bounds on All Attributes	**TRACKING ID DISPATCH:** reference_code + Email Placeholders	**INTEGRITY RULES:** Foreign Keys + Cascade Deletions

1. Table: users

Authentication credentials and authorization records for clinic medical staff and administrators.

Column Name	Data Type (Length)	Null ?	Constraints	Default	Description & Purpose
`id`	`INTEGER (11)`	NO	PK, AUTOINCREMENT	None	Unique internal user ID.
`email`	`VARCHAR (150)`	NO	UNIQUE	None	Official EARIST institutional email for login.
`password_hash`	`VARCHAR (255)`	NO	None	None	Bcrypt salted hash (cost factor 10) for secure login.
`role`	`VARCHAR (20)`	NO	CHECK('assistant','admin')	None	Authorization role: 'assistant' (Doctor/Nurse), 'admin'.
`first_name`	`VARCHAR (100)`	NO	None	None	Given name of clinic practitioner.
`last_name`	`VARCHAR (100)`	NO	None	None	Surname of clinic practitioner.
`id_number`	`VARCHAR (30)`	YES	None	NULL	Employee/Staff ID (e.g. MED-2026-008).
`contact_number`	`VARCHAR (20)`	YES	None	NULL	Mobile contact number or clinic extension.
`created_at`	`DATETIME (19)`	YES	None	CURRENT_TIMESTAMP	User account registration timestamp.

2. Table: assistants

Medical staff and doctor profile extensions (1-to-1 relationship with users).

Column Name	Data Type (Length)	Null ?	Constraints	Default	Description & Purpose
`assistant_id`	`INTEGER (11)`	NO	PK, AUTOINCREMENT	None	Unique primary key for doctor profile.
`user_id`	`INTEGER (11)`	NO	FK, UNIQUE -> users(id)	None	Foreign key referencing users table (ON DELETE CASCADE).
`specialization`	`VARCHAR (150)`	YES	None	NULL	Clinical medical specialization (General Physician, Dental, etc.).
`room_number`	`VARCHAR (50)`	YES	None	NULL	Assigned physical station (e.g. Main Clinic Rm 101).
`is_on_duty`	`TINYINT (1)`	YES	CHECK (0 or 1)	1	Active duty flag: 1 = Currently on duty, 0 = Off duty.
`is_active`	`TINYINT (1)`	YES	CHECK (0 or 1)	1	Active roster status flag: 1 = Active, 0 = Inactive.

3. Table: doctor_schedules

Weekly template schedules defining regular consultation shifts for medical practitioners.

Column Name	Data Type (Length)	Null ?	Constraints	Default	Description & Purpose
schedule_id	INTEGER (11)	NO	PK, AUTOINCREMENT	None	Unique recurring schedule window identifier.
assistant_id	INTEGER (11)	NO	FK -> assistants	None	Foreign key referencing assistants(assistant_id).
day_of_week	VARCHAR (15)	NO	None	None	Day of clinic duty (Monday to Friday).
start_time	VARCHAR (10)	NO	None	None	Duty shift start time in 24-hr format (e.g. 08:00).
end_time	VARCHAR (10)	NO	None	None	Duty shift end time in 24-hr format (e.g. 12:00).
max_slots	INTEGER (5)	YES	None	15	Maximum patient capacity allowed per duty window.
is_available	TINYINT (1)	YES	CHECK (0 or 1)	1	Schedule toggle: 1 = Active/Bookable, 0 = Inactive.

4. Table: doctor_custom_availability

Date-specific schedule overrides (leaves, seminars, holiday blackouts, modified hours).

Column Name	Data Type (Length)	Null ?	Constraints	Default	Description & Purpose
id	INTEGER (11)	NO	PK, AUTOINCREMENT	None	Unique primary key for schedule override.
assistant_id	INTEGER (11)	NO	FK -> assistants	None	Foreign key referencing assistants(assistant_id).
date	VARCHAR (10)	NO	UNIQUE(asst, date)	None	Specific calendar date formatted as YYYY-MM-DD.
is_available	TINYINT (1)	YES	CHECK (0 or 1)	1	0 = Doctor on leave / unavailable; 1 = Custom hours.
custom_times	VARCHAR (255)	YES	None	NULL	JSON array/string defining custom slot hours.
notes	TEXT (500)	YES	None	NULL	Administrative explanation (e.g. Official Training / Leave).
created_at	DATETIME (19)	YES	None	CURRENT_TIMESTAMP	Override record creation timestamp.

5. Table: appointments

Central consultation records, triage, diagnosis, prescriptions, and email tracking placeholders.

Column Name	Data Type (Length)	Null ?	Constraints	Default	Description & Purpose
appointment_id	INTEGER (11)	NO	PK, AUTOINCREMENT	None	Unique primary key for appointment.
reference_code	VARCHAR (25)	NO	UNIQUE	None	**Tracking ID (Reference Code):** Issued to patient (e.g. EAR-2026-A109) to track their appointment progress online.
patient_type	VARCHAR (20)	YES	CHECK('student','faculty_staff')	'student'	Campus constituent category: 'student' or 'faculty_staff'.
patient_name	VARCHAR (150)	NO	None	None	Full legal name of the patient.
id_number	VARCHAR (30)	NO	None	None	Student Number (e.g. 234-0000M) or Faculty Employee ID.
department	VARCHAR (150)	NO	None	None	College (e.g. College of Engineering) or Administrative Office.
course	VARCHAR (150)	YES	None	NULL	Academic degree program (BSCpE, BSIT, etc., nullable for staff).
year_section	VARCHAR (20)	YES	None	NULL	Year level & section (e.g. 3-B).
age	INTEGER (3)	YES	None	NULL	Patient age in completed years.
gender	VARCHAR (10)	YES	None	NULL	Gender identity ('Male', 'Female', 'Other').
address	VARCHAR (255)	YES	None	NULL	Residential / dormitory address in Metro Manila.
contact_number	VARCHAR (20)	NO	None	None	Primary contact mobile number for notifications & triage.
patient_email	VARCHAR (150)	YES	**[NEW PLACEHOLDER]**	NULL	**Patient Email Address:** Target email address where Tracking ID & appointment receipt are sent.
symptoms	TEXT (1000)	YES	None	NULL	Self-reported symptoms and health complaints.
symptom_severity	VARCHAR (15)	YES	CHECK('Mild','Moderate','Severe')	'Mild'	Preliminary triage priority rating.
allergies	TEXT (255)	YES	None	NULL	Known drug or food allergies (e.g. Penicillin, Aspirin).
medical_history	TEXT (1000)	YES	None	NULL	Pre-existing chronic medical history or prior treatments.
assistant_id	INTEGER (11)	NO	FK -> assistants	None	Foreign key referencing attending doctor.
appointment_date	VARCHAR (10)	NO	None	None	Scheduled consultation date formatted as YYYY-MM-DD.
time_slot	VARCHAR (30)	NO	None	None	Consultation time slot window (e.g. 09:00 AM - 09:30 AM).
service_type	VARCHAR (100)	NO	None	None	Clinical service requested (General Consultation, Dental, Clearance).
status	VARCHAR (20)	YES	CHECK('Pending','Approved','OnGoing','Completed','Cancelled')	'Approved'	Consultation workflow lifecycle state.
email_status	VARCHAR (15)	YES	CHECK('Pending','Sent','Failed') **[NEW]**	'Pending'	**Tracking Email Dispatch Status:** Tracks whether tracking ID was delivered to patient's email.
email_sent_at	DATETIME (19)	YES	**[NEW PLACEHOLDER]**	NULL	**Tracking Email Timestamp:** Date and time the tracking notification email was delivered.
reason	TEXT (500)	YES	None	NULL	Patient's stated reason for campus clinic visit.
diagnosis	TEXT (1000)	YES	None	NULL	Official medical diagnosis recorded by attending physician.
prescription	TEXT (1000)	YES	None	NULL	Prescribed medications and pharmaceutical orders.
dosage_instructions	TEXT (1000)	YES	None	NULL	Directions for medication administration and self-care.
created_at	DATETIME (19)	YES	None	CURRENT_TIMESTAMP	Initial appointment submission timestamp.

6. Table: email_notifications [NEW OUTBOX DISPATCH LOG]

Audit log and outbox queue tracking all automated confirmation emails and tracking code dispatches.

Column Name	Data Type (Length)	Null ?	Constraints	Default	Description & Purpose
notification_id	INTEGER (11)	NO	PK, AUTOINCREMENT	None	Unique primary key for email dispatch log.
appointment_id	INTEGER (11)	NO	FK -> appointments	None	Foreign key referencing parent appointment (ON DELETE CASCADE).
recipient_email	VARCHAR (150)	NO	None	None	Recipient email address receiving the tracking link and ticket.
subject	VARCHAR (255)	NO	None	None	Email subject line (e.g. EARIST Clinic Confirmation - Ref: EAR-2026-A109).
reference_code	VARCHAR (25)	NO	None	None	The tracking reference code embedded in the message body.
template_name	VARCHAR (50)	YES	None	'appointment_tracking_id'	Email template identifier for body rendering.
status	VARCHAR (15)	YES	CHECK('Pending','Sent', 'Failed')	'Pending'	Delivery status: 'Pending' in outbox, 'Sent', or 'Failed'.
error_message	TEXT (1000)	YES	None	NULL	Error diagnostics or SMTP server response if delivery failed.
sent_at	DATETIME (19)	YES	None	NULL	Exact timestamp when email was transmitted.
created_at	DATETIME (19)	YES	None	CURRENT_TIMESTAMP	Notification queue record creation timestamp.

Referential Integrity & Operating Business Rules

- **Foreign Key Enforcement:** SQLite PRAGMA foreign_keys = ON; is initialized on every connection.
- **Data Type Length Constraints:** Attributes adhere to industry-standard maximum storage limits (e.g. VARCHAR(150) for institutional emails, VARCHAR(25) for tracking IDs, TEXT(1000) for clinical triage and diagnoses).
- **Cascade Deletions:** Deleting a medical practitioner cascades to all schedules, availability overrides, and appointments; deleting an appointment purges corresponding outbox notifications.
- **Globally Unique Tracking ID:** reference_code in appointments is enforced unique; formatted as EAR-XXXX-XXX ensuring conflict-free patient process tracking.
- **Capacity Throttling:** Maximum capacity of 5 patients per consultation slot (MAX_SLOT_CAPACITY = 5) ensures high clinical care standards and eliminates queue congestion.
- **Duplicate Booking Safeguard:** Patients are prevented from scheduling overlapping appointments on identical dates and time windows.
- **Email Outbox Processing:** The email_notifications table acts as an asynchronous outbox queue. Future background jobs (cron or worker) poll rows with status = 'Pending', dispatch via SMTP/Nodemailer, and mark both email_notifications.status = 'Sent' and appointments.email_status = 'Sent' with timestamps.